Research Report

A Comprehensive Treatment Approach for Patellofemoral Pain Syndrome in Young Women

Background and Purpose. The purposes of this study were (1) to evaluate a comprehensive treatment approach for patients with patellofemoral pain syndrome and (2) to compare a training program using isometric muscle contractions with a training program using eccentric muscle contractions. **Subjects.** Forty female patients with patellofemoral pain syndrome, aged 15 to 28 years ($\bar{X}$=20.2, SD=3.2), were randomly assigned to either a group using isometric muscle contractions or a group using eccentric muscle contractions. **Methods.** The effects of 12 weeks of treatment, consisting of an educational component and a training program, on physical activity, pain, and muscle function were evaluated after 3 and 12 months. **Results.** No differences were found between the two groups, except in one of the torque measurements. A reduction in pain and improvements in torque, vertical jumping ability, and physical activity level were seen in both groups after treatment. At the 12-month follow-up, 85% of the subjects were participating in sports without pain and 37 subjects rated their overall knee function as excellent or good. **Conclusion and Discussion.** The results indicate that the improvements shown in this study may be due to spontaneous recovery over time, the education given to the subjects, the pain monitoring system, the gradually progressing training program, and the adjusted physical activity. [Thomeé R. A comprehensive treatment approach for patellofemoral pain syndrome in young women. *Phys Ther.* 1997;77:1690–1703.]

Key Words: *Gradually progressing training program, Pain monitoring system, Physical activity, Standardized information, Visual analog scale.*

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Patellofemoral pain syndrome (PFPS) consists of anterior knee pain excluding intra-articular pathology, peripatellar tendinitis, and bursitis[1] and is one of the most common knee problems among physically active adolescents and young adults.[2–10] Surgical treatment is rarely indicated,[11–13] and the most frequently recommended treatment for PFPS is exercise.[12–15] The effects of exercise programs on PFPS, however, are not well documented.

Patellofemoral pain syndrome seems to be caused by the overloading of the patellofemoral joint during physical activity rather than by malalignment of the lower extremity.[1,6,16–19] Pain from the patellofemoral joint and reduced or altered physical activity[1,6,7,12–14,16,18–21] result in decreased force production by the knee extensors,[6,7,9,22–28] predominantly during eccentric contractions,[26–28] with evidence of selective disturbances within the quadriceps femoris muscle, particularly at certain knee angles.[27–32] Authors often advocate the use of isometric muscle contractions[22,33] as well as eccentric muscle contractions,[26,27] but recently there have been recommendations for vastus medialis muscle training and functional lower-extremity training.[10,15] The importance of including an educational component in treatment programs has also been emphasized.[1,14]

In a previous study of lower-extremity muscle function in patients with PFPS,[28] my colleagues and I found decreased vertical jumping ability when subjects used the most symptomatic leg compared with the least symptomatic leg. In addition, we found lower isometric, concentric, and eccentric isokinetic knee extensor torque and electromyographic (EMG) activity in patients with PFPS compared with age- and gender-matched control subjects in the range closer to full knee extension and differences in EMG activity between the vastus medialis and rectus femoris muscles. We believe that these results were due to inhibition of muscle activity from the patellofemoral joint, an inhibition that was more pronounced at knee angles closer to full knee extension and to the vastus medialis muscle. In another study,[32] my colleagues and I reinforced our findings by superimposing, on the quadriceps femoris muscle, maximally tolerated single-twitch electrical stimulation on maximal voluntary knee extensor activation in a sitting position. In that study, 10 of the 11 patients with PFPS demonstrated increased torque as compared with 2 of the 9 control subjects. These findings indicate a decreased ability, estimated at 18%, among patients with PFPS to maximally contract the quadriceps femoris muscle isometrically. Thus, the treatment approach in the present study was based in part on the premise that the inhibition of quadriceps femoris muscle activity is a symptom leading to weakness.

The aims of this study were (1) to evaluate the effects of a comprehensive treatment approach consisting of an educational component and a training program on physical activity level, pain, and muscle function in a

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well-defined group of patients with PFPS having long-standing pain that had not improved with periods of rest or traditional treatment and (2) to compare a treatment program consisting of predominantly isometric contractions with a treatment program consisting of predominantly eccentric contractions.

Method

Subjects

Local orthopedic surgeons referred patients who, for a minimum of 6 months, had unsuccessfully tried periods of no treatment and no physical activity, as well as some form of therapy, after contacts with orthopedic surgeons, school physicians, school nurses, or physical therapists. The patients were included in the study if three of the following four inclusion criteria were fulfilled: (1) They experienced pain from the patellofemoral joint during or after activity, (2) they had pain from the patellofemoral joint during or after sitting, (3) they had pain from the patellofemoral joint during stair climbing, and (4) they had pain from the patellofemoral joint during squatting.

The exclusion criteria that eliminated subjects from participation in the study were:

- a history of any recurrent patellar subluxation or dislocation
- a history of intermittent or persistent knee joint swelling during the previous year
- other injuries to the knee joint, such as any tears of the menisci, ligaments, or joint capsule
- known damage to the articular cartilage
- overuse symptoms of tendinitis, bursitis, medial plica, or synovitis during the previous year
- any major muscle or tendon ruptures in the lower extremities
- surgery carried out in the lower extremities

Forty women with PFPS with a mean symptom duration of 43 months (SD=31.2, range=6–108) participated in the study. Only women were chosen for this study because PFPS is considered to be more common in women than in men and because its etiology can be different between men and women.[1,3,7,20] The subjects had a mean age of 20.2 years (SD=3.2, range=15–28), a mean weight of 64.1 kg (SD=8.8, range=50–85), and a mean height of 169 cm (SD=6.4, range=156–180). Informed consent was obtained from all subjects. The mean period of no treatment and no physical activity was 8.4 months (SD=2.7, range=4–15). In a previous study,[19] in which the same 40 subjects participated, measurements of lower-extremity alignment and measurements of lower-extremity range of motion (ROM) indicated that there were no differences between the subjects' right and left sides. The following anthropometric and ROM measurements were evaluated (for a detailed methodological description, see Thomeé et al[19]):

- quadriceps femoris muscle angle (Q-angle) in a standing position at 0 and 30 degrees of knee flexion
- distance between anterior superior iliac spines
- distance between the most medial parts of the knees
- lower-extremity length
- knee hyperextension angle
- angles between the lower leg and the calcaneus, the calcaneus and the horizontal plane, and the lower leg and the horizontal plane
- arch index
- congruence angle, patellar tilt angle at an angle of 0 and 20 degrees of knee flexion, and patellar subluxation at an angle of 0 and 20 degrees of knee flexion, as measured radiographically
- ROM for hip flexion, extension, abduction, and medial (internal) and lateral (external) rotation; knee flexion; and ankle dorsiflexion

In a previous study,[19] my colleagues and I found that the 40 subjects with PFPS who participated in the present study did not differ in lower-extremity alignment, patellar alignment, or lower-extremity ROM measurements when compared with a group of 20 age- and gender-matched control subjects without a history of pain disorders. To assess reliability, all anthropometric and ROM measurements were documented from videotaped recordings by two different examiners, who were unaware of each other's results and of the subjects' identity. In addition, measurements were documented twice by the same examiner, who was unaware of the subjects' identity.[19] For anthropometric and ROM measurements, the errors of measurement (expressed as a percentage of the mean value) ranged from 2% to 6%. When two examiners were compared, the errors of measurement ranged from 4% to 8%.

Power calculations of sample size showed that my acceptance of the null hypothesis was safe (ie, sample size was sufficient) regarding no difference between the subjects' knees and between subjects with PFPS and control subjects. One standard deviation was chosen as the least acceptable difference between groups. All calculations showed, for an alpha level of .05, a beta error of less than 0.20 and a power of greater than 0.80.

The subjects' knees were referred to as the "most symptomatic knee" and the "least symptomatic knee" because 27 subjects (68%) complained of symptoms from both knees. The subjects, however, reported no impact on the

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ability to work or physical activity due to symptoms from the least symptomatic knee. Using "pain from the patellofemoral joint during the last 6 months resulting in absence of at least 1 day from school, work, or physical activity" as a criterion for disablement, all 40 subjects in the study could have been classified as unilaterally affected.

Physical activity level, pain, muscle function, and overall knee function were evaluated before treatment and at 3 months and 12 months after treatment. All tests were administered by the same investigator and at the same facility.

Documentation of Physical Activity Level

The subjects' participation in sports activities, despite pain, during the 12 months preceding the study was documented, along with the frequency of activity. To assess at what pain-free activity level the subjects could perform, Tegner and Lysholm's activity rating system,[34] originally designed and reliability tested for knee ligament injuries, was used. The subjects were asked at which activity level they could perform without experiencing pain from the patellofemoral joint during or after activity. This score was originally constructed for knee ligament injuries and classifies activities into 10 different ranked categories according to competitive sports, recreational sports, and work.

Documentation of Pain

A questionnaire,[19] originally designed based on construct validity and not tested for reliability, was used to assess pain ("yes" or "no") experienced from the patellofemoral joint in the most symptomatic knee in nine different situations:

- at rest after activity
- during heavy loading of the patellofemoral joint, such as during high-intensity sports activities or weight training and lifting or carrying near-maximal weights
- after 10 minutes of submaximal loading of the patellofemoral joint, such as during low-intensity sports activities or weight training and lifting or carrying submaximal weights
- during walking up 1, 2, and 3 or more sets of stairs
- during walking down 1, 2, and 3 or more sets of stairs
- during squatting
- during sitting with knees flexed at a 90-degree angle for less than 30 minutes and for 30 minutes or longer
- after sitting with knees flexed at a 90-degree angle for less than 30 and for 30 minutes or longer
- during jogging 2 to 5 km

A more detailed assessment of pain experienced from the patellofemoral joint was made with a visual analog scale[35] (VAS) (0–10) in 12 different situations (items). A VAS was chosen because it has been proven to be both reliable and appropriate for measuring pain.[36] Zero on the VAS used in this study reflects "no pain," and 10 reflects "pain as bad as it could be."[35,37] The subjects were asked to make marks on a 10-cm horizontal line. Results were documented to the nearest millimeter. The first three items were maximal level of pain, minimal level of pain, and average daily maximal level of pain experienced during the last 3 months. After 12 weeks of treatment, the subjects reported pain experienced during the final 2 weeks of treatment. For the next four items, pain was provoked by palpation of the patella and by a compression test. For these two tests, a constant force of approximately 35 N was used.[19] The force was standardized by pressing on a Penny and Giles myometer (model D60 107 MK4)* that digitally showed the pressure exerted (in newtons). The examiner pressed repeatedly with his index finger on the myometer in the same manner as done in the palpation and compression tests described below, until 35 N of force could be easily reproduced.

For the palpation test, the most painful spot of the patella was palpated with the tip of the index finger. Each subject made marks on the VAS for the most symptomatic knee and for the least symptomatic knee. During the compression test, the examiner pressed his index finger superior to the patella and perpendicular to the femur, with the same constant force as described above. The subject was then asked to slowly contract the quadriceps femoris muscle. Pain from the most symptomatic knee and from the least symptomatic knee was registered with the VAS. The five remaining items related to pain experienced during different physical activities. The subjects were asked to walk up three sets of stairs, totaling 90 steps, each 17 cm in height, and mark the level of pain they experienced on the VAS and then walk down the same stairs and again mark the level of pain experienced on the VAS. Next, the subjects were asked to report pain experienced during isometric knee extension torque measurements in a sitting position, concentric and eccentric knee extension torque measurements in a sitting position, and a vertical jump test.

The VAS scores, which should be regarded as data on an ordinal scale,[35,38] were transformed to interval-scale data using a Rasch analysis.[39] A Rasch pain measure was calculated from the VAS scores for all 12 items for each subject, and the significance of changes in individual subjects during treatment was tested.[40]

* Penny and Giles Transducers, Christchurch, England.

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Documentation of Muscle Function

For the strength tests, a Kin-Com® II dynamometer[†] (software version 4.01 C, 1987) was used. After 5 minutes of bicycling on a Monark ergometer[‡] at 100 W of resistance, the subject was positioned with a hip angle of 120 degrees in the testing chair. The knee joint axis was approximated to the Kin-Com® axis. The leg was secured to the Kin-Com® shin pad at 3 cm proximal to the insertion of the tibialis anterior muscle with the ankle joint at 90 degrees. The trunk, hip, and thigh were strapped down to avoid unwanted movements. For all tests, individual lower-leg gravity correction was used. Standardized acceleration (medium) and preloading start forces (50 N) were used for all subjects for all tests.

Muscle activity was recorded using a full-wave rectified low-pass filtered EMG unit with a 60-Hz notch filter at a sampling frequency of 100 Hz and a bandwidth of 72 to 490 Hz. The raw EMG signal was monitored on an oscilloscope (Techtronix 2200[§]) for control of artifacts.

The following measurements were obtained for both knees[28]: (1) isometric knee extension torque measurements at a knee angle of 60 degrees, (2) concentric and eccentric knee extension torque measurements at an angular velocity of 30°/s between knee angles of 90 degrees to 5 degrees (average torques across the ranges of 25°–50° and 50°–75° were documented), (3) recordings of EMG activity from the vastus medialis and rectus femoris muscles during torque measurements, and (4) measurements recorded during a vertical jump test.

The pretreatment torque measurements and EMG recordings were taken twice, on two separate days, with 2 days between, for 15 patients.[28] There were no systematic differences (Wilcoxon nonparametric test, $P>.05$) between test days for the torque or EMG measurements. The average methodological error was 5.6% and 9.9% for the torque and EMG measurements, respectively.

After skin preparation (ie, shaving, scraping, and cleaning with pure alcohol), the EMG electrodes (Red Dot model 2239 monitoring electrodes[‖]) were secured over the vastus medialis and rectus femoris muscles.[28] The positions of the EMG electrodes were standardized by positioning the subject in the dynamometer as described with a knee angle of 60 degrees and drawing a straight line from the medial joint line at the medial part of the patellar tendon to the anterior superior iliac spine. For the vastus medialis muscle, an ink dot was placed on the line at 25% of the distance from the medial joint line. The first electrode was placed on the dot, and the second electrode was placed on the line distal to the dot at an interelectrode distance of 2 cm. For the rectus femoris muscle, an ink dot was placed on the line at 66% of the distance from the medial joint line. The first electrode was placed on the dot, and the second electrode was placed on the line distal to the dot at an interelectrode distance of 2 cm. The ground electrode was placed on the bony part of the medial part of the knee.

Electromyographic data recorded from the vastus medialis and rectus femoris muscles during concentric and eccentric muscle contractions were normalized to EMG activity obtained during maximal isometric contractions at a 60-degree angle of knee flexion and expressed as a percentage of that value.

Isometric Testing

After submaximal isometric contractions for further warm-up and familiarization, three to four trials of maximal isometric contractions, with a 1-minute rest between trials, were done at a 60-degree angle of knee flexion. Torque and EMG recordings were measured as an average of a 2-second interval for each trial.

Isokinetic Testing

Only one speed (30°/s) was chosen for the isokinetic testing in order not to overuse the subjects' knee and because no speed-specific results were expected. After five to six submaximal concentric and eccentric contractions at the angular velocity of 30°/s, each subject performed three to four maximal trials. Each trial consisted of a concentric contraction from 90 degrees to 5 degrees of knee extension, followed, after a 30-second rest interval, by an eccentric contraction from 5 degrees to 90 degrees of knee extension. A 1-minute rest was given between trials. The torque and EMG activity were averaged over the ranges of 25 to 50 degrees and 50 to 75 degrees of knee extension from the trial with the highest peak torque. These two ranges were chosen for torque and EMG activity to cover the early and late parts of knee extension.

The ranges of 25 to 50 degrees and 50 to 75 degrees were chosen for both concentric and eccentric contractions even though these actions are in opposite directions. The same order of testing was used for each subject both before treatment and at the 3- and 12-month follow-ups.

Vertical Jump Test

A vertical jump test was performed on a different day than the torque measurements.[28] I used this test because it combines eccentric and concentric contractions and because it requires a high speed of muscle contraction. Each subject was placed on a wooden platform. A belt was secured around the subject's waist, and a tape

[†] Chattanooga Group Inc, 4747 Adams Rd, PO Box 489, Hixson, TN 37343.
[‡] Quinton Fitness Equipment, 3303 Monte Villa Pkwy, Bothell, WA 98021.
[§] Techtronix Inc, 26600 SW Parkway, PO Box 1000, Wilsonville, OR 97070.
[‖] 3M Medica, Borken, Westfalen, Germany.

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measure ran from the belt vertically down through a loop in the platform. When the subject performed a vertical jump, the tape measure was pulled through the loop and the height was recorded. After warm-up and familiarization with several submaximal jumps, three different types of jumps were performed, with three maximal trials for each jump and a 1-minute rest between the jumps:

1. A bilateral jump from a static self-chosen squatted position (STATIC). The subjects' arms were kept crossed on the back.

2. A bilateral counter-movement drop jump from a 20-cm-high wooden box (DROP 20). The subjects stood on the edge of the box and were instructed to fall down to the platform and jump up vertically as high as possible. The subjects' arms were allowed to assist freely.

3. A unilateral counter-movement drop jump (DROP 20-UL) on right and left legs, respectively, as in the DROP 20 jump.

The best result for each type of jump was recorded to the nearest centimeter.

Documentation of Overall Knee Function

At the 3- and 12-month follow-ups, the subjects were instructed, in writing, to summarize their overall knee function, including symptoms, muscle function, and physical activity level, as "excellent," "good," or "poor."

Treatment Program

The subjects were numbered consecutively as they were included in the study and then assigned to either a group using isometric contractions (subjects with odd numbers) or a group using eccentric contractions (subjects with even numbers). A detailed, comprehensive treatment program was developed, consisting of an educational component and a training program. In the educational component, education was given about (1) the general concepts of PFPS, including incidence, etiology, and available treatment, (2) basic patellofemoral anatomy and physiology, using anatomic diagrams illustrating lower-extremity skeletal parts, major muscles and their functions, and how nutrition is provided for articular cartilage, (3) pain and symptoms specific for PFPS, and (4) pain, physical activity, and the use of the VAS. The 12-week training program consisted of (1) lower-extremity training exercises (Fig. 1), (2) a detailed progression plan (Fig. 2), (3) a pain monitoring system (Fig. 3), and (4) a diary for documentation of training results and symptoms.

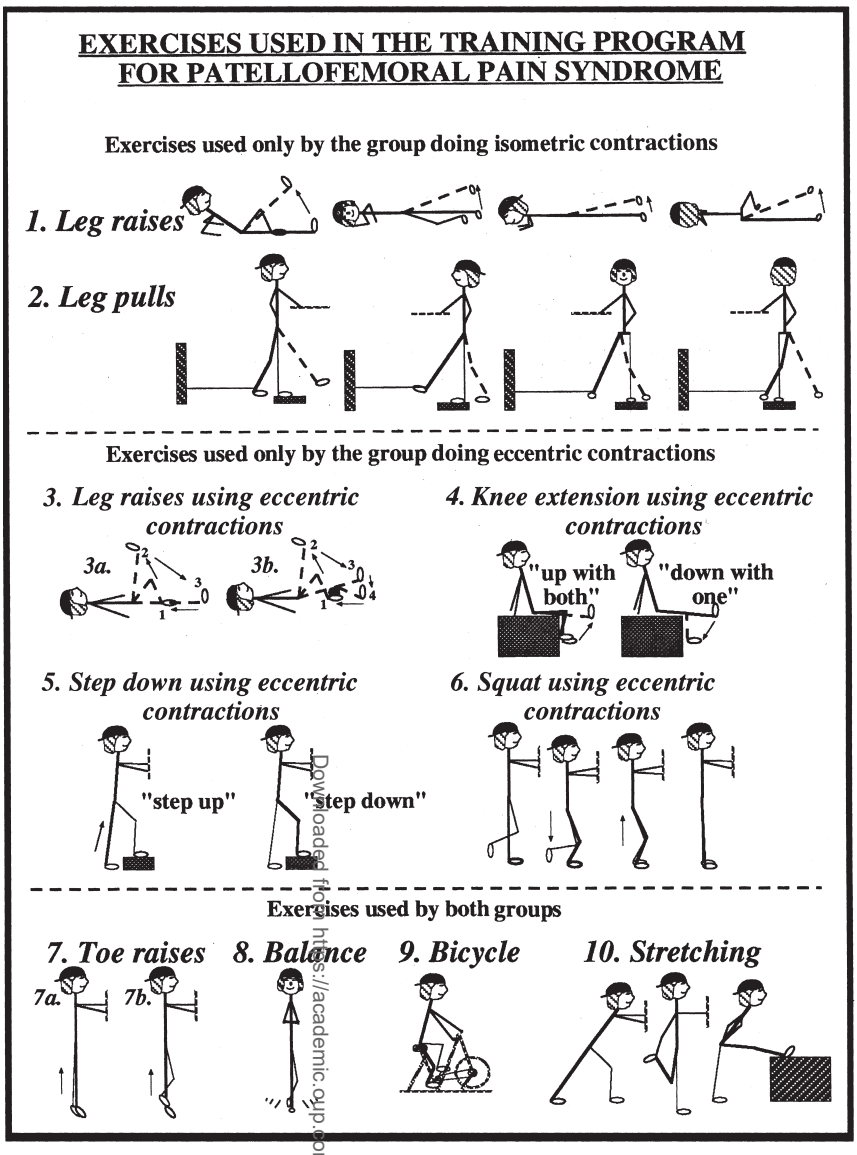

Figure 1.
Exercises followed by the isometric and eccentric contraction groups.

Three sessions were used for education and to familiarize the subjects with the training program. Twelve weeks of treatment then started, with daily training for the first 2 weeks to further accustom the subjects to all the details of the program. During the first 2 weeks, each subject was seen by the physical therapist three times a week to allow for adjustments of the training program and to allow the subjects to ask questions. After the first 2 weeks, the frequency of training was three times per week and

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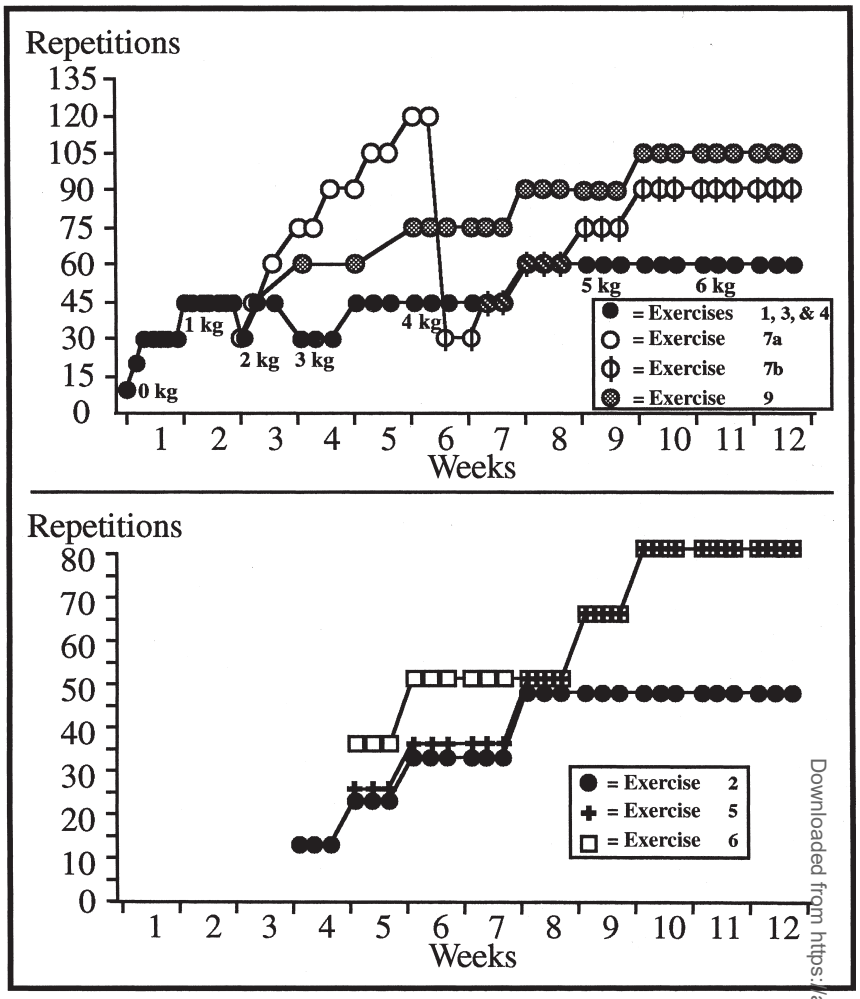

Figure 2.
Progression plan (based on author's clinical experience) for the number of repetitions used in the exercises (illustrated in Fig. 1) followed by the isometric and eccentric contraction groups. The loads (in kilograms) used in exercises 1, 3, and 4 are indicated. For exercise 9, the number of repetitions should be multiplied by a factor of 10.

each subject was seen by the physical therapist once or twice every 2 weeks.

Training program. The group using isometric contractions performed four different straight leg raises (exercise 1) during the first 2 weeks. Beginning with the third week, toe raises on both feet (exercise 7a) were added as well as stretching of the knee extensors, flexors, and plantar flexors (exercise 10). At the start of the fourth week, exercises in a standing position were included (ie, hip flexion, extension, adduction, and abduction with a rubber tube [exercise 2] along with an exercise for lower-extremity balance [exercise 8] and bicycling on a stationary bicycle [exercise 9]). Toe raises on one foot (exercise 7b) were included at week 6. For all exercises, except when bicycling and stretching knee extensors, the subjects were instructed not to flex their knees.

For the group using eccentric contractions, two different leg-lowering exercises (exercises 3a and 3b) were used, along with doing eccentric contractions in a sitting position (exercise 4). From the fourth week, step downs and one-legged squats using eccentric contractions (exercises 5 and 6) were added, along with an exercise for lower-extremity balance (exercise 8) and bicycling on a stationary bicycle (exercise 9). Toe raises were done on one foot (exercise 7b) from week 6.

Correct technique for achieving the full range and control of the speed of movement was emphasized for all exercises. After 6 weeks, the increased load of the training program required a pretraining warm-up consisting of the same exercises as used in the training program, but with approximately one third of the weight and number of repetitions. Hip exercises (flexion, extension, adduction, and abduction), toe raises, balance exercises, bicycling, and stretching exercises were performed bilaterally. The other exercises were performed only on the most symptomatic side. The subjects were encouraged to use a stationary bicycle at home or to bicycle outdoors on level ground according to the treatment program. Ankle weights, rubber tubes, and knee rolls (exercise 3b) were supplied at no cost to the subjects.

The progression of exercises and load were individually prescribed using the progression plan (Fig. 2), the pain monitoring system (Fig. 3), and the diary. The diary provided for the subjects was used for motivational purposes and for documentation of training results and symptoms. The information was used to reach an agreement on how further treatment should progress.

Pain monitoring system. A pain monitoring system was used in which pain and discomfort during the rehabilitation period were documented using the VAS. Pain experienced momentarily during any exercise was allowed up to a VAS pain level of 5, although only in the case that the pain subsided immediately after the exercise was completed. Visual analog scale pain levels were chosen based on my clinical experience. I considered pain reported up to a level of 2 to be "safe," pain up toa level of 5 to be "acceptable," and pain above level 5 to be

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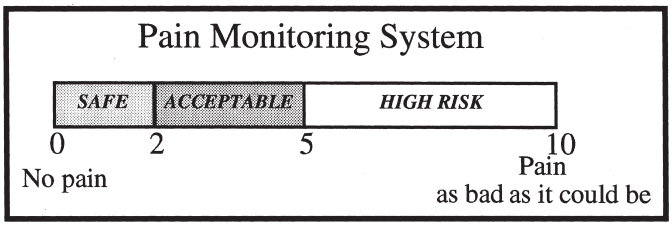

Figure 3.
Visual analog scale for pain monitoring from 0 (no pain) to 10 (pain as bad as it could be) with safe, acceptable, and high-risk pain zones indicated.

"high risk" (Fig. 3). I also considered pain after the program up to a level of 5 to be "acceptable," but all pain should have subsided by the next morning. If pain did not subside, the level of the exercise program was lowered one step. The subjects were allowed to continue their participation in physical activities during the treatment period using the pain monitoring system as it was used in the training program.

Data Analysis

Conventional methods were used for the calculation of medians, means, and standard deviations. The Wilcoxon nonparametric signed-rank test was used to compare the subjects' most symptomatic knee with the least symptomatic knee and to evaluate effects of treatment within groups. The Mann-Whitney nonparametric *U* test was used to compare the two groups. For comparisons of frequencies of subjects with symptoms and frequencies of subjects engaged in sports, Fisher's exact test was used for differences between the two groups and a nonparametric sign test was used for differences within groups. All computations were done on the StatView 4.5 statistical package.[#]

Results

All 40 patients completed their treatment program. From analyses of the subjects' diaries, it was estimated that approximately 90% of the training and physical activity during the treatment period was performed in the safe zone of the pain monitoring system (ie, below a VAS pain level of 2). Three subjects from each group continued their program, supervised by a physical therapist, for a mean period of 6 weeks (range=4–12) after the primary 12 weeks of treatment. Two subjects, one from each group, who were not satisfied with their progress chose to undergo a lateral retinacular release at approximately 9 months after the initial treatment started and were asymptomatic during sports activities at the 12-month follow-up. Data from all subjects were included in the statistical analysis and are included in the presented figures. At the 12-month follow-up, 37 subjects rated their knee function as being excellent or good, and 1 subject rated her knee function as improved but still poor. The two subjects who underwent surgery rated their knee function as excellent.

[#] Abacus Concepts Inc, 1984 Bonita Ave, Berkeley, CA 94704.

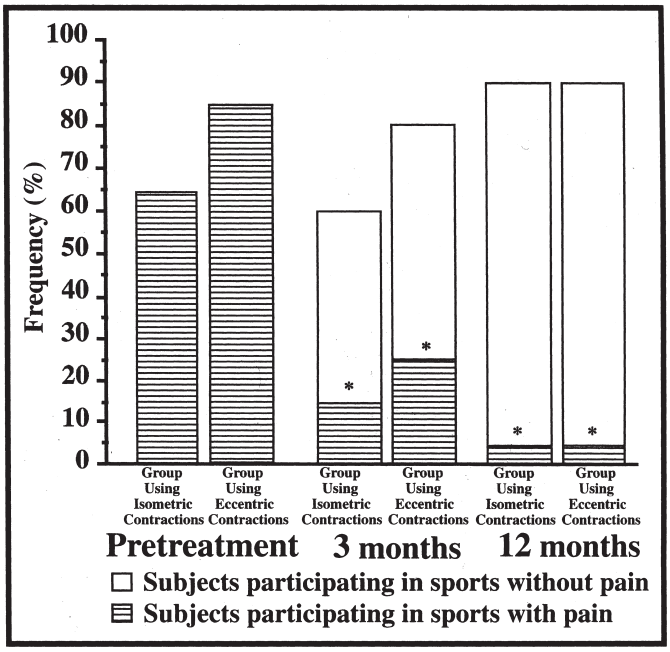

Figure 4.
Frequency of subjects in the isometric and concentric contraction groups participating in competitive or recreational sports. The dashed portions of the bars indicate subjects with pain, and the white portions of the bars indicate subjects without pain. Asterisk (*) indicates significant reduction ($P<.05$) compared with pretreatment assessment.

Physical Activity Level

There was no change in the number of subjects (n=30) who participated in sports after 3 months of treatment, but fewer of these subjects (n=8) ($P<.01$) experienced pain during sports activities at the 3-month follow-up as compared with before treatment. At the 12-month follow-up, 90% of the subjects (n=36) in both groups were participating in sports and only 5% (n=2) of all subjects reported pain during their sports activities. No differences were found between the two exercise groups (Fig. 4).

The pain-free activity level[34] increased after 3 months of treatment ($P<.01$) and between 3 and 12 months post-treatment ($P<.05$) in both exercise groups, with no difference between the groups. At the 12-month follow-up, the median pain-free activity level was 4.0 (interquartile range=2.0).

Pain

A reduction in the number of subjects experiencing pain from the patellofemoral joint in the most symptomatic knee was seen in all nine situations used in the questionnaire at both 3- and 12-month follow-ups ($P<.01$). There was no difference between the two groups. At the 12-month follow-up, all subjects tolerated jogging 2 km

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Table.
Mean, Standard Error (SE), and Range of Knee Extensor Torque (in Newton-meters) During Isometric (Maximal), Concentric, and Eccentric (Average Between Knee Angles of 25°–50° and 50°–75°) Contractions at Pretreatment Assessment and at 3- and 12-Month Follow-ups for the Group Using Isometric Contractions and the Group Using Eccentric Contractions

	Isometric (60°)		Concentric				Eccentric			
			25°–50°		50°–75°		25°–50°		50°–75°	
	Group Using Isometric Contractions	Group Using Eccentric Contractions	Group Using Isometric Contractions	Group Using Eccentric Contractions	Group Using Isometric Contractions	Group Using Eccentric Contractions	Group Using Isometric Contractions	Group Using Eccentric Contractions	Group Using Isometric Contractions	Group Using Eccentric Contractions
Pretreatment										
$\bar{X}$	130.6	151.6	87.0	100.8	113.8	133.2	104.8	121.7	122.1	147.7
SE	6.8	8.0	4.9	4.6	4.9	5.8	7.5	7.2	8.6	8.2
Range	81–200	82–210	43–125	74–133	69–147	91–171	48–186	47–174	58–215	77–212
3-mo follow-up										
$\bar{X}$	147.6	161.5	100.0	112.4	130.3	144.5	122.6	136.9	141.0	159.5
SE	7.3	7.7	5.0	5.6	6.3	7.0	5.8	7.6	6.9	10.1
Range	89–209	97–229	44–143	81–167	55–171	95–225	57–164	82–187	80–209	88–273
12-mo follow-up										
$\bar{X}$	154.1	175.1	106.6	116.4	136.0	147.9	128.9	159.4	144.1	180.9
SE	7.0	7.5	3.9	4.9	5.0	6.6	5.3	5.2	5.9	7.4
Range	87–209	121–239	76–140	80–160	93–170	99–226	83–186	94–178	101–194	130–267

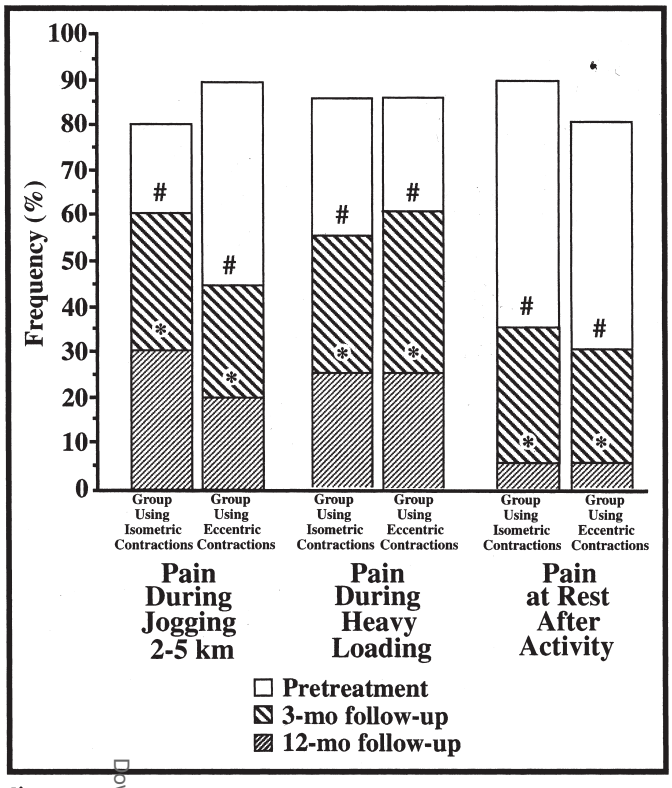

Figure 5.
Frequency of subjects with pain in their most symptomatic knee during jogging 2 to 5 km; during heavy loading of the patellofemoral joint, such as during high-intensity sports activities or weight training and lifting or carrying near-maximal weights; and at rest after activity for the isometric and eccentric contraction groups before treatment and at 3 and 12 months after treatment. Number sign (#) indicates significant reduction ($P<.01$) compared with pretreatment assessment. Asterisk (*) indicates significant reduction ($P<.01$) compared with 3-month follow-up.

and only 10 (25%) of the 40 subjects complained of pain during jogging 2 to 5 km or during heavy loading of the patellofemoral joint. Two subjects (5%) complained of pain at rest after activity (Fig. 5).

There was a reduction in the scores on the VAS in all 12 items, both at the 3-month follow-up and between 3 and 12 months posttreatment ($P<.01$). A significance test of the overall Rasch pain measurement calculated for each subject from all 12 items revealed a reduction in pain at 3 months posttreatment (Fig. 6a), compared with the pretreatment measurement, in 23 of the 40 subjects and at 12 months posttreatment (Fig. 6b) in all subjects ($P<.05$). There were no differences between the two groups.

Muscle Function

Results of the measurements of torque for knee extension during isometric, concentric, and eccentric contractions are presented in the Table. Both groups had increased torque in the most symptomatic knee at the 3- and 12-month follow-ups as compared with the pretreat-

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ment assessment ($P<.05$) (Fig. 7). No differences were found between the two groups, except for a smaller increase in average torque during eccentric contractions across the range of 25 and 50 degrees at the 3-month follow-up in the group using eccentric contractions compared with the group using isometric contractions. The average increases in torque for the two groups were, at the 3-month follow-up, 10% during isometric contractions, 12% during concentric contractions, and 13% during eccentric contractions. The increases in torque were 17%, 17%, and 24%, respectively, at the 12-month follow-up as compared with the pretreatment assessment. No increases were found in the subjects' least symptomatic knee, and no difference in torque was found between the subjects' least symptomatic and most symptomatic knees at the 12-month follow-up.

There was no difference in average EMG activity, normalized to maximal isometric activation, of the vastus medialis or rectus femoris muscle for knee angles of 25 to 50 degrees or 50 to 75 degrees from the pretreatment evaluation to the 12-month follow-up either between the subjects' least symptomatic and most symptomatic knees or between the two groups.

The difference in EMG activity between the two ranges (50°–75°) and (25°–50°) was calculated. The difference was reduced during concentric contraction of the knee extensors for the vastus medialis muscle ($P<.01$), but not for the rectus femoris muscle, at the 3-month follow-up compared with the pretreatment assessment. A reduction during eccentric contraction of the knee extensors was seen for the vastus medialis muscle, but not for the rectus femoris muscle, at both at the 3- and 12-month follow-ups (Fig. 8).

Increases were seen in vertical jumping ability in the bilateral jumps (STATIC and DROP 20) ($P<.01$). The increase was 4% at the 3-month follow-up and 5% between the 3- and 12-month follow-ups. No differences were found between the two groups. The subjects increased their vertical jumping ability ($P<.01$) with the most symptomatic leg from a mean of 24.1 cm (SD=7.8, range=4–41) before treatment to 27.0 cm (SD=6.2, range=16–40) at the 3-month follow-up and to 28.7 cm (SD=5.7, range=18–40) at the

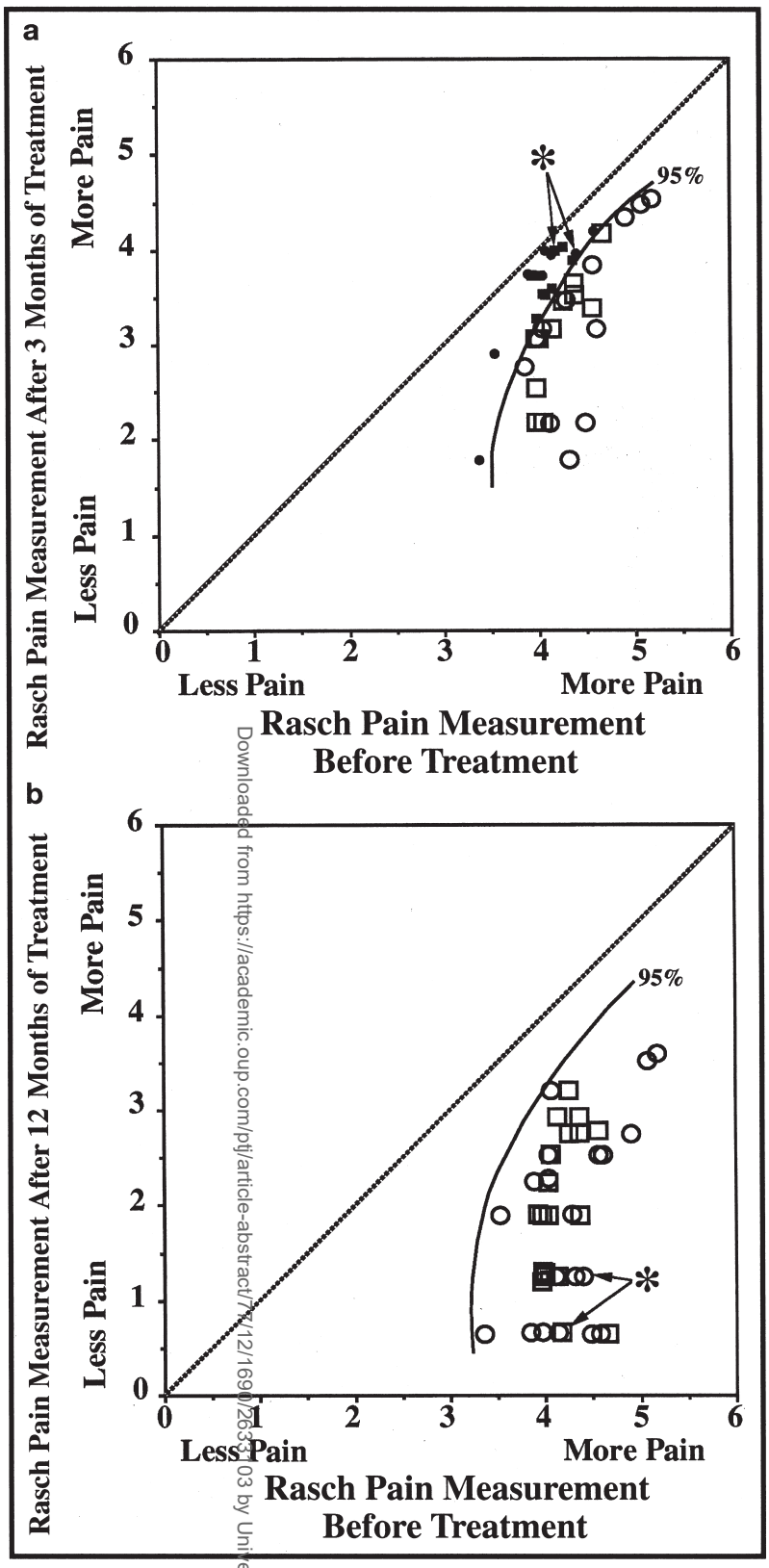

Figure 6.
Rasch pain measurement calculated for each subject from all 12 items at the 3-month follow-up (a) and the 12-month follow-up (b), as compared with the pretreatment assessment, with identity line and 95% quality (95% confidence interval) line. Subjects whose Rasch pain measurement improved significantly ($P<.05$) are identified with nonfilled symbols. Asterisk (*) indicates the two subjects who underwent surgery at 9 months after the initial treatment started.

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12-month follow-up. With the least symptomatic leg, the subjects jumped 26.2, 28.5, and 29.2 cm before treatment, at the 3-month follow-up, and at the 12-months follow-up, respectively, with a difference between pretreatment assessment and the 12-month follow-up ($P<.01$). There was no difference between the subjects' least symptomatic and most symptomatic legs from the pretreatment assessment to the 12-month follow-up.

Discussion

My study examined a treatment approach for PFPS in young women and compared a treatment program consisting predominantly, but not exclusively, of isometric contractions with a treatment program consisting predominantly, but not exclusively, of eccentric contractions. The approach was based on education, a pain monitoring system, a training program, and a progression plan allowing for the individually designed treatment approach advocated in the literature.[1,12–15,41–43] No differences were found between the two groups, except in one of the torque measurements. A reduction in pain and improvements in torque and physical activity level were seen in both groups after 12 weeks of treatment. Furthermore, all 40 subjects in this study had less pain, as analyzed with the Rasch analysis, at the 12-month follow-up as compared with before treatment. At the 12-month follow-up, 85% of the patients participated in competitive or recreational sports without pain and 37 subjects rated their knee function as excellent or good.

The reasons for improvement by all of the subjects may be an effect of time, the education, the pain monitoring system, the gradually progressing training program, and the adjusted physical activity. In my study, there was no control group that did not receive treatment, which limits the possibility of evaluating the effects of treatment. The subjects, however, to some extent can be seen as their own controls, having an average of 8 months of no treatment or physical activity without relief of symptoms. Thus, I believe that time is not likely to be a major reason for the improvement. It is likely that the education resulted in a better understanding of PFPS and that the information given to the subjects may have contributed to an altered physical activity pattern. The subjects were allowed to continue, using the pain monitoring system, with adjusted physical activity during treatment. The pain monitoring system resulted, for example, in an avoidance or reduction of temporary heavy loading of the patellofemoral joint.

No differences in activity level, pain, or muscle function were found between the two groups, except for a smaller increase in torque during eccentric contractions in the knee angle range of 25 to 50 degrees for the group using eccentric contractions compared with the group using isometric contractions. Thus, effects of treatment do not seem to be sensitive to a particular choice of exercise in the training program. A reduction in pain was reported by most subjects after a few minutes of training. Improved muscle activity with increased motor unit recruitment as well as a more normal activity of the

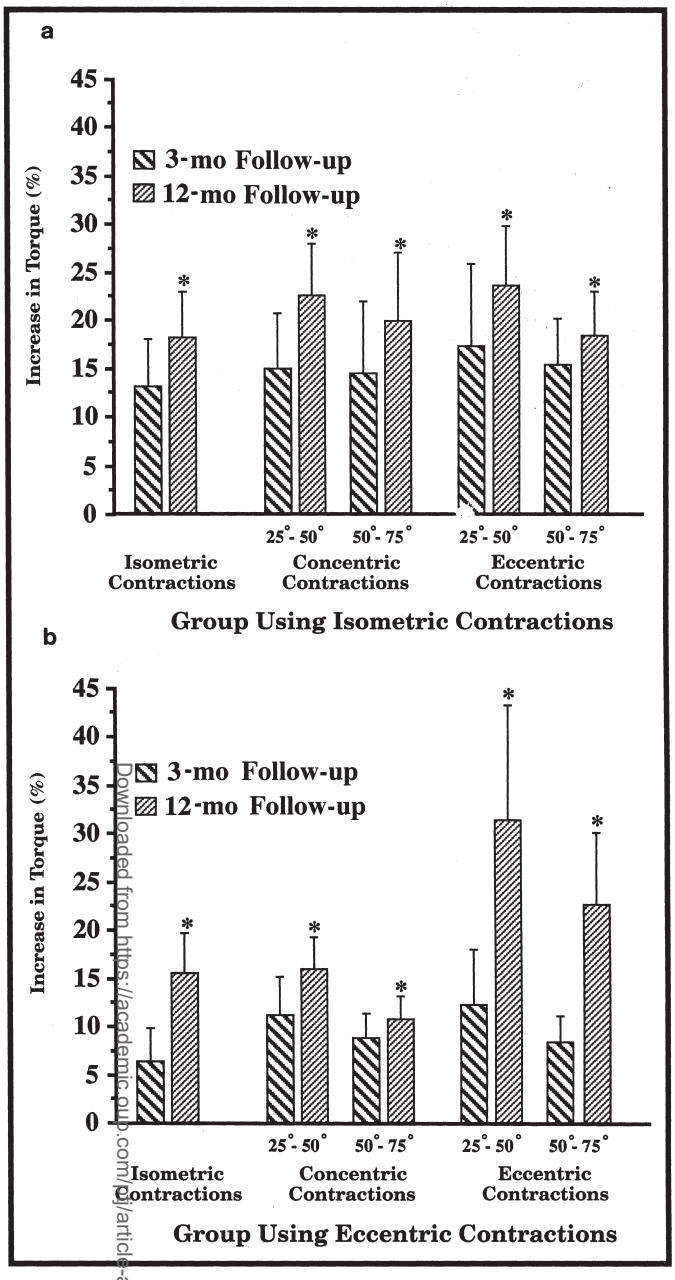

Figure 7.
Increase in torque during isometric, concentric, and eccentric contractions in the subjects' most symptomatic knee at the 3- and 12-month follow-ups, compared with the pretreatment assessment, for the group using isometric contractions (a) and the group using eccentric contractions (b). All increases were significant ($P<.05$) compared with pretreatment values, but there were no differences between the two groups except for a smaller increase ($P<.05$) in torque during eccentric contractions in the knee angle range of 25 to 50 degrees at the 3-month follow-up in the group using eccentric contractions compared with the group using isometric contractions. Asterisk (*) indicates significant difference ($P<.05$) between the 3- and 12-month follow-ups.

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vastus medialis muscle theoretically could be achieved when pain is decreased. In my study, a decrease in EMG activity between the two ranges (50°–75°) and (25°–50°) for the vastus medialis muscle at the 12-months follow-up compared with the pretreatment evaluation (Fig. 8) suggests improved muscle activity. The difference between the ranges at the 12-month follow-up, however, was still higher ($P<.05$) compared with a group of 20 subjects with no known history of PFPS studied by Thomeé et al[28] with the same apparatus and methods as in the present study, indicating that full normalization of vastus medialis muscle activity was still not present in the patients with PFPS at the 12-month follow-up.

Increases in torque during isometric, concentric, and eccentric contractions in the most symptomatic knee at 12 months posttreatment resulted in no difference in torque between the subjects' knees. I believe that the training program caused positive effects on patellofemoral joint structures.[45–48] The adaptive changes can possibly be explained by increased diffusion of nutrients to the cartilage caused by loading and unloading on the patellofemoral joint[3,44] and by improved nutrition of surrounding joint structures and muscles caused by the increased blood circulation. The need for a carefully controlled training program seems to be important because exercises that are too strenuous or biomechanically unfavorable may cause degeneration of the musculoskeletal tissue, with articular cartilage being the most sensitive tissue.[48] The increased physical activity after treatment may contribute to the further reduction in symptoms and improvement in muscle function seen at the 12-month follow-up.

Taping for patellar malalignment[49] was not considered for use in the present study because subjects had no signs of patellar malalignment, as evaluated with computed tomography.[19] A recent study[50] demonstrated poor to fair reliability among 12 physical therapists comparing a series of patellofemoral alignment tests to determine when and how patellofemoral taping techniques should be used. This finding was not surprising because patellar malalignment measured clinically under static conditions is not a common characteristic of PFPS.[1,6,17,51]

With the VAS scores, analyzed with the Rasch analysis, pain levels were assessed as they related to different pain-provoking situations. The changes in individual subjects during rehabilitation were examined, and I identified subjects showing different recovery times. The pain monitoring system was used, and pain during and after exercises or activity was allowed. This approach is in contrast to recent articles on treatment of patellofemoral pain that advocate that all training should be done pain-free.[1,14,41–43] It is my experience that using a level of

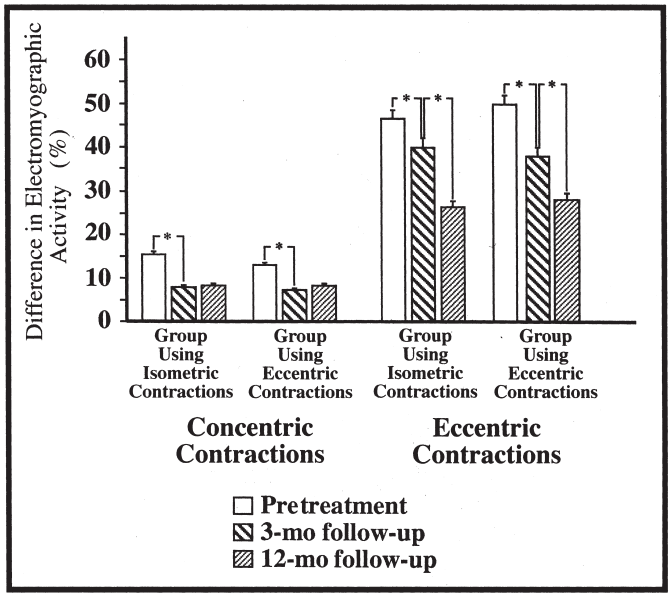

Figure 8.
Difference in electromyographic activity of the vastus medialis muscle of the subjects' most symptomatic knee between the two ranges of knee extension (50°–75° and 25°–50°) for the isometric and eccentric contraction groups at the pretreatment assessment and at the 3- and 12-month follow-ups. Asterisk (*) indicates significance level of $P<.01$.

no pain during or after activity as a criterion for increasing physical activity and loads during treatment for patients with PFPS requires such a decrease in physical activity and loads that improvements can fail to appear. An accepted pain level that is too high, however, can result in patient frustration. The pain monitoring system used in my study allowed for a gradual exercise progression, with minimized risks for overloading and overuse of the patellofemoral joint. Because patient adherence was very high with no dropouts, I advocate the use of the pain monitoring system for treatment of patients with PFPS. Further research is needed using a control group receiving no treatment to properly evaluate the effects of the treatment approach used in this study.

Conclusions

No differences were found in treatment effects between the two groups studied, except in one of the torque measurements. The treatment approach used in this study seems to be successful, but further study of this approach with a proper control group is needed. After treatment, the subjects experienced less pain and showed improved activity levels and muscle function. Reasons for improvement may be found in spontaneous recovery over time, the education given, the pain monitoring system, the gradually progressing training program, and the adjusted physical activity. Training programs for patients with PFPS primarily focus on monitoring pain and gradually progressing load and repetitions of lower-extremity exercises seem to be warranted.

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References

1 Reid DC. The myth, mystique, and frustration of anterior knee pain. *Clinical Journal of Sports Medicine.* 1993;3:139–143.

2 Newman D, Thomson J, Barnes J, Moore T. A clinic for athletic injuries. *Proc R Soc Med.* 1969;62:939–941.

3 James S. Chondromalacia of the patella in the adolescent. In: Kennedy JC, ed. *The Injured Adolescent Knee.* Baltimore, Md: Williams & Wilkins; 1979:205–251.

4 Clement DB, Taunton JE, Smart GW, McNicol KL. A survey of running injuries. *The Physician and Sportsmedicine.* 1981;9:47–58.

5 Devereaux MD, Lachmann SM. Athletes attending a sports injury clinic: a review. *Br J Sports Med.* 1983;17:137–142.

6 Fairbank JCT, Pynsent PB, Van Poortvliet JA, Phillips H. Mechanical factors in the incidence of knee pain in adolescents and young adults. *J Bone Joint Surg Br.* 1984;66:685–693.

7 Kujala UM, Österman K, Kvist M, et al. Factors predisposing to patellar chondropathy and patellar apicitis in athletes. *Int Orthop.* 1986;10:195–200.

8 Kannus P, Järvinen M, Niitymäki S. Computerized recording of visits to an outpatient sports clinic. *Am J Sports Med.* 1987;15:79–85.

9 MacIntyre DL, Robertson GE. Quadriceps muscle activity in women runners with and without patellofemoral pain syndrome. *Arch Phys Med Rehabil.* 1992;73:10–14.

10 O'Neill DB, Micheili LJ, Warner JP. Patellofemoral stress: a prospective analysis of exercise treatment in adolescents and adults. *Am J Sports Med.* 1992;20:151–156.

11 Sandow MJ, Goodfellow JW. The natural history of anterior knee pain in adolescents. *J Bone Joint Surg Br.* 1985;67:36–38.

12 Zappala FG, Taffel CB, Scuderi GR. Rehabilitation of the patellofemoral joint. *Orthop Clin North Am.* 1992;23:555–566.

13 Tria AJ Jr. Conservative care for patellar pain. *Orthop Clin North Am.* 1992;23:545–554.

14 Shelton GL, Thigpen LK. Rehabilitation of patellofemoral dysfunction: a review of literature. *J Orthop Sports Phys Ther.* 1991;14:243–249.

15 Doucette SA, Goble EM. The effect of exercise on patellar tracking in lateral patellar compression syndrome. *Am J Sports Med.* 1992;20:434–440.

16 Werner S. An evaluation of knee extensor and knee flexor torques and EMGs in patients with patello-femoral pain syndrome in comparison with matched controls. *Knee Surgery, Sports Traumatology, and Arthroscopy.* 1995;3:89–94.

17 Milgrom C, Kerem E, Finestone A, et al. Patellofemoral pain caused by overactivity: a prospective study of risk factors in infantry recruits. *J Bone Joint Surg Am.* 1991;73:1041–1043.

18 Galanty HL, Matthews C, Hergenroeder AC. Anterior knee pain in adolescents. *Clinical Journal of Sports Medicine.* 1994;4:176–181.

19 Thomeé R, Renström P, Karlsson J, Grimby G. Patellofemoral pain syndrome in young women, I: a clinical analysis of alignment, common symptoms, and functional activity level. *Scandinavian Journal of Medicine and Science in Sports.* 1995;5:237–244.

20 Fulkerson JP, Hungerford DS. *Disorders of the Patellofemoral Joint.* Baltimore, Md: Williams & Wilkins; 1990.

21 Chesworth BM, Matthews C, Hergenroeder AC. Validation of outcome measure in patients with patellofemoral pain syndrome. *J Orthop Sports Phys Ther.* 1989;11:302–308.

22 Kannus P, Niitymäki S. Which factors predict outcome in the nonoperative treatment of patellofemoral pain syndrome? A prospective follow-up study. *Med Sci Sports Exerc.* 1993;26:289–296.

23 McMullen W, Roncarati A, Koval P. Static and isokinetic treatments of chondromalacia patellae: a comparative investigation. *J Orthop Sports Phys Ther.* 1990;12:256–266.

24 Ficat RP, Philippe J, Hungerford DS. Chondromalacia patellae: a system of classification. *Clin Orthop.* 1979;144:55–62.

25 Bentley G, Dowd G. Current concepts of etiology and treatment of chondromalacia patellae. *Clin Orthop.* 1984;189:209–228.

26 Bennett JG, Stauber WT. Evaluation and treatment of anterior knee pain using eccentric exercise. *Med Sci Sports Exerc.* 1986;18:526–530.

27 Werner S, Eriksson E. Isokinetic quadriceps training in patients with patellofemoral pain. *Knee Surgery, Sports Traumatology, and Arthroscopy.* 1993;1:162–168.

28 Thomeé R, Renström P, Karlsson J, Grimby G. Patellofemoral pain syndrome in young women, II: muscle function in patients and healthy controls. *Scandinavian Journal of Medicine and Science in Sports.* 1995;5:245–251.

29 Möller BN, Krebs B, Tideman-Dal C, Aaris K. Isometric contractions in the patellofemoral pain syndrome: an electromyographic study. *Arch Orthop Trauma Surg.* 1986;105:24–27.

30 Souza DR, Gross MT. Comparison of vastus medialis obliquus: vastus lateralis muscle integrated electromyographic ratios between healthy subjects and patients with patellofemoral pain. *Phys Ther.* 1991;71:310–320.

31 Voight ML, Wieder DL. Comparative reflex response times of vastus medialis obliquus and vastus lateralis in normal subjects and subjects with extensor mechanism dysfunction. *Am J Sports Med.* 1991;19:131–137.

32 Thomeé R, Grimby G, Svantesson U, Österberg U. Quadriceps muscle performance in sitting and standing in young women with patellofemoral pain syndrome and healthy controls. *Scandinavian Journal of Medicine and Science in Sports.* In press.

33 DeHaven KE, Dolan WA, Mayer PJ. Chondromalacia patellae in athletes. *Am J Sports Med.* 1979;7:5–11.

34 Tegner Y, Lysholm J. Rating systems in the evaluation of knee ligament injuries. *Clin Orthop.* 1985;198:43–49.

35 Scott J, Huskisson EC. Graphic representation of pain. *Pain.* 1976;6:175–184.

36 Price DD, McGrath PA, Rafii A, Buckingham B. The validation of visual analogue scales as ratio scale measures for chronic and experimental pain. *Pain.* 1983;1:45–56.

37 Seymour RA, Simpson JM, Charlton JE, Philips ME. An evaluation of length and end-phrase of visual analogue scales in dental pain. *Pain.* 1985;21:177–185.

38 Chapman SL, Casey KL, Dubner R, et al. Pain measurement: an overview. *Pain.* 1985;22:1–31.

39 Wright BD, Stone MH. *Best Test Design.* Chicago, Ill: Mesa Press; 1979.

40 Thomeé R, Grimby G, Wright BD, Linacre JM. Rasch analysis of visual analog scale measurements before and after treatment of patients with patellofemoral pain syndrome. *Scand J Rehabil Med.* 1995;27:145–151.

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41 Beckman M, Craig R, Lehman RC. Rehabilitation of the patellofemoral dysfunction in the athlete. *Clin Sports Med.* 1989;8:841–860.

42 Brunet ME, Stewart GW. Patellofemoral rehabilitation. *Clin Sports Med.* 1989;8:319–329.

43 Gray D. Physical therapy techniques for conservative treatment of patellar pain and instability. *Adaptive Techniques in Sports Medicine.* 1994;2:263–272.

44 Curtiss PH. Changes produced in the synovial membrane and synovial fluid by disease. *J Bone Joint Surg Am.* 1964;46:873–888.

45 Wright W, Dowson D, Seller P. Bio-engineering aspects of synovial fluid and cartilage. *Modern Trends in Rheumatology.* 1971;2:21–29.

46 Radin EL. The physiology and degeneration of joints. *Semin Arthritis Rheum.* 1973;2:245–257.

47 Paukkonen K, Selkäinaho K, Jurvelin J, et al. Cells and nuclei of articular cartilage chondrocytes in young rabbits enlarged after nonstrenuous exercise. *J Anat.* 1985;142:13–20.

48 Kannus P, Jozsa L, Renström P, et al. The effects of training, immobilization, and remobilization on musculoskeletal tissue. 1: training and immobilization. *Scandinavian Journal of Medicine and Science in Sports.* 1992;2:100–118.

49 McConnell J. The management of chondromalacia patellae: a long-term solution. *Australian Journal of Physical Therapy.* 1986;32:215.

50 Fitzgerald GK, McClure PW. Reliability of measurements obtained with four tests for patellofemoral alignment. *Phys Ther.* 1995;75:84–92.

51 Eng JJ, Pierrynowski MR. Evaluation of soft foot orthotics in the treatment of patellofemoral pain syndrome. *Phys Ther.* 1993;73:62–70.

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